Thermal Hydraulics for the Concentric Reactor Model

Modelica loop and fuel heat-structure note

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Contents

1 Purpose

This note documents the thermal-hydraulic model now used for the Reactor Kinetics Lab concentric reactor design. The model is not a licensing-grade subchannel calculation. It is a compact Modelica plant model intended to provide:

- primary-loop inlet and outlet coolant temperatures,
- primary mass flow and core pressure drop diagnostics,
- fuel and moderator effective temperatures for reactivity feedback,
- an effective D_2O moderator density for density feedback,
- a transient fuel heat capacity between fission power and coolant heat removal.

The key change from the earlier annular-core note is that the thermal model is now framed around the OpenMC concentric fuel element. The hydraulic circuit remains a single effective primary channel, but the fuel heat storage and heat-transfer area are derived from the resolved concentric fuel-ring geometry.

2 Concentric Reactor Inputs

The first-pass thermal model uses the OpenMC concentric fuel-element defaults as its core geometry source of truth:

Quantity	Value
Nominal thermal power P	20 MW
Fuel ring count	6
Fuel meat thickness	1.5 cm
Coolant gap between rings	4.5 cm
Fuel-element inner radius	20 cm
Fuel-element outer radius	60 cm
Active fuel height	350 cm
Fuel density	12.2 g/cm^3
Primary coolant/moderator	D_2O represented by the current water approximation

From these values, the aggregate fuel-meat quantities are

$$V_f \approx 0.7917 \text{ m}^3, \qquad m_f \approx 9.66 \times 10^3 \text{ kg}, \qquad A_{ht} = \frac{V_f}{t_f/2} \approx 105.6 \text{ m}^2,$$

where $t_f = 1.5$ cm is the fuel meat thickness. The implied average fuel-meat power density at 20 MW is approximately

$$q_f''' = \frac{P}{V_f} \approx 25.3 \text{ MW/m}^3.$$

The effective Modelica core pipe uses the OpenMC active height. Its diameter is intentionally retained as a calibrated single-channel hydraulic diameter, not the 1.2 m fuel-envelope outer diameter. The concentric OpenMC dimensions enter the fuel heat storage and heat-transfer area directly; pressure drop remains a coarse loop diagnostic rather than a resolved subchannel prediction.

3 Operating Envelope

The nominal primary-loop operating point is unchanged:

Quantity	Selected value
Primary inlet bulk temperature target	25°C
Nominal outlet bulk temperature target	45°C
Nominal bulk temperature rise	20 K
Nominal mass flow	237 kg/s
Core flow direction	top-to-bottom
Primary pressure regime	low pressure, single phase

At this stage the pump remains flow-controlled at the nominal mass flow. The inlet plenum is represented as an open expansion volume, which supplies an absolute pressure reference without a bidirectional pressure-source side branch. The pressure drop is therefore a diagnostic for the current effective-pipe geometry, not a validated subchannel loss prediction. In particular, the current pipe hydraulic diameter is an effective calibration value used to keep the primary-loop pressure model well conditioned.

4 Core Thermal Model

4.1 Coolant channel

The primary loop represents the active region as one axially discretised Modelica dynamic pipe. It carries the full primary mass flow top-to-bottom and exposes one thermal connector per axial node. With ideal heat-port coupling, each heat port temperature is the local bulk coolant temperature used by the pipe energy balance.

The detailed distribution of flow among concentric gaps is not modelled. The current model therefore should not be interpreted as a hot-channel or margin-to-boiling calculation.

4.2 Fuel heat structure

Each axial coolant node is coupled to an aggregated RELAP-style one-dimensional fuel half-slab. The slab represents the path from the fuel meat symmetry plane to the coolant interface:

$$\left.\frac{\partial T_f}{\partial x}\right|_{x=0} = 0, \qquad q''_j = h_{core}\left(T_{wall,j} - T_{cool,j}\right).$$

The half-slab is discretised into $N_r = 5$ solid nodes. For axial node j and radial solid node r the Modelica model uses:

- `HeatCapacitor fuelNode[j,r]` for fuel heat storage,
- `PrescribedHeatFlow fuelHeat[j,r]` for deposited fission power,
- `ThermalConductor` links between adjacent radial fuel nodes,
- a half-cell wall conductor from the outer fuel node to the wall,
- `Convection coreConvection[j]` from the wall to the coolant heat port.

The heat-transfer coefficient is currently fixed,

$$h_{core} = 2.0 \times 10^4 \ \mathrm{W/(m^2\,K)},$$

so that the v1 heat path is tunable and independent of the effective pipe hydraulic area. Dittus-Boelter or a custom D_2O correlation can replace this fixed coefficient later, but that should be done with explicit hydraulic constants for the concentric coolant gaps.

4.3 Feedback temperatures

The fuel feedback temperature is a power-weighted axial average of the radial fuel-node average:

$$T_{f,eff} = \sum_{j=1}^{N_z} f_j \left(\frac{1}{N_r} \sum_{r=1}^{N_r} T_{f,j,r} \right),$$

where f_j are the normalised axial power fractions supplied to the Modelica model.

The moderator/coolant feedback temperature is the same axial weighting applied to the bulk coolant temperatures:

$$T_{m,eff} = \sum_{j=1}^{N_z} f_j T_{cool,j}.$$

In the current Modelica implementation $T_{cool,j}$ is taken from `core.heatPorts[j].T`. If a future toolchain exposes alias issues with those heat ports, the fallback approximation is

$$T_{m,eff} \approx \frac{T_{in} + T_{out}}{2}.$$

The moderator density feedback uses a compact D_2O density approximation rather than the current light-water medium density:

$$\rho_{D2O}(p, T) = \rho_{ref} \left[1 - \beta(T - T_{ref}) + \kappa(p - p_{ref})\right],$$

with $\rho_{ref} = 1105$ kg/m^3, $T_{ref} = 300$ K, $p_{ref} = 4.0 \times 10^5$ Pa, $\beta = 3.0 \times 10^{-4}$ K^{-1}, and $\kappa = 4.5 \times 10^{-10}$ Pa^{-1}. The effective SI density is power-weighted over the same axial nodes:

$$\rho^{SI}_{m,eff} = \sum_{j=1}^{N_z} f_j \rho_{D2O}(p_j, T_j), \qquad \rho_{m,eff} = \frac{\rho^{SI}_{m,eff}}{1000}.$$

Here $\rho^{SI}_{m,eff}$ is in kg/m^3 and $\rho_{m,eff}$ is in g/cm^3 for neutronics feedback interfaces that expect cgs density.

The model also exposes diagnostic maximum temperatures:

- `T_fuelCenterlineMax`, the maximum inner fuel-node temperature,
- `T_fuelWallMax`, the maximum fuel wall temperature.

5 Modelica Interface

The FMI-facing variables are:

Direction	Signal	Description
Python → Modelica	`totalPower`	total fission power [W]
Python → Modelica	`axialPowerFractions[n]`	normalised axial power fractions
Modelica → Python	`T_inlet`	core inlet bulk temperature [K]
Modelica → Python	`T_outlet`	core outlet bulk temperature [K]
Modelica → Python	`massFlow`	primary mass flow rate [kg/s]
Modelica → Python	`dp_core`	core pressure drop [Pa]
Modelica → Python	`T_fuelEff`	Doppler-relevant effective fuel temperature [K]
Modelica → Python	`T_moderatorEff`	effective moderator/coolant temperature [K]
Modelica → Python	`rho_m_eff_SI`	effective D_2O moderator density [kg/m^3]
Modelica → Python	`rho_m_eff`	effective D_2O moderator density [g/cm^3]
Modelica → Python	`T_fuelCenterlineMax`	diagnostic peak fuel centerline node [K]
Modelica → Python	`T_fuelWallMax`	diagnostic peak fuel wall [K]

The initial conditions are set near the nominal 20 MW operating point to avoid starting coupled point-kinetics runs from cold fuel. This is still an approximate initialization, not a formal steady-state solve.

6 Deliberate Limits

The current implementation deliberately excludes:

- ring-by-ring or gap-by-gap coolant flow distribution,
- hot-channel factors and safety-margin calculations,
- temperature-dependent fuel properties,
- D_2O-specific medium replacement in Modelica,
- decay heat and shutdown cooling,
- active pool cooling and detailed reflector thermal hydraulics,
- pump coastdown and full pump head-curve dynamics.

These omissions keep the first coupled neutronics/thermal-hydraulics model small enough to verify while still providing the feedback temperatures needed for Doppler and moderator reactivity terms.